

Reliability Ranking Classroom Run Kit

TOK Cognitive Science Activity Bank • Cross-AOK • 25-35 min

Category	Cross-AOK	Time	25-35 min
Difficulty	Medium	ID	reliability-ranking

Big idea: Certainty varies across kinds of knowledge.

One-Minute Teacher Brief

Students compare certainty, usefulness and interpretation across knowledge claims.

Student Prompt

Inspect Reliability Ranking Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Reliability Ranking Stimulus A	Student-facing stimulus 1: Knowledge claims ranked by certainty, usefulness, and risk if wrong. Name the AOK, the method being used, and the standard of evidence you would apply.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Reliability Ranking Stimulus B	Student-facing stimulus 2: A second ranking after students learn the method behind each claim. Name the AOK, the method being used, and the standard of evidence you would apply.	Use this for comparison, a second round, or a partner challenge.
Reliability Ranking Reveal / Twist	Student-facing stimulus 3: A discussion on whether reliability means the same thing in every AOK. Name the AOK, the method being used, and the standard of evidence you would apply.	Teacher reveal: connect the changed response to the big idea — Certainty varies across kinds of knowledge.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Reliability Ranking Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Reliability Ranking Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Is certainty the best measure of knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Cross-AOK.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how certainty varies across kinds of knowledge.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Is certainty the best measure of knowledge?

Setup Checklist

- Prepare a starter stimulus using: Knowledge claims ranked by certainty, usefulness, and risk if wrong.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- Knowledge claims ranked by certainty, usefulness, and risk if wrong.
- A second ranking after students learn the method behind each claim.
- A discussion on whether reliability means the same thing in every AOK.

Resources To Use

- Resource pack: <resources/reliability-ranking/index.html>
- Card deck CSV: <resources/reliability-ranking/cards.csv>
- Visual prompt: <resources/reliability-ranking/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Give cards such as: "The angles in a Euclidean triangle sum to 180 degrees", "Smoking increases lung cancer risk", "This poem expresses grief", "Most people conform under group pressure", "This painting is beautiful".
2. Students rank claims from most to least certain.
3. They then rank from most to least useful.
4. Compare the two rankings.

Debrief Moves

- Knowledge question: Is certainty the best measure of knowledge?
- Can uncertain knowledge still be valuable?
- Are mathematical truths more valuable because they are more certain?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...